

# File Permissions & File Search

9 of 16 Lab Exercises - Linux Fundamentals

| Exercises Completed | Success Rate | Environment     |
|---------------------|--------------|-----------------|
| 9 / 16              | 56%          | Rocky Linux 9.7 |

## Summary & Overview

Completed 9 practical exercises focused on Linux file permissions and file search operations. These exercises demonstrate understanding of **chmod syntax**, **find command variations**, and **permission filtering**. This hands-on training in Rocky Linux 9.7 provides foundational knowledge required for LFCS and RHCSA certifications, with emphasis on file system security and troubleshooting capabilities essential for NOC-to-SysAdmin career advancement.

## Completed Exercises

| Ex. | <b>Topic</b>               | <b>Command / Skill</b>         | <b>Status</b> |
|-----|----------------------------|--------------------------------|---------------|
| 1   | File Search by Time        | find /dev/ -mmin -5            | ✓             |
| 2   | Permission Removal         | chmod g-w                      | ✓             |
| 3   | Permission Filtering       | find with -perm /g+w           | ✓             |
| 4   | File Search by Size        | find /home/bob/ -size          | ~             |
| 6   | Find & Redirect Output     | find /usr/share -name dogs.txt | ✓             |
| 7   | File Copy with Permissions | sudo cp file /opt/             | ✓             |
| 8   | Find Directories           | find /var/ -type d             | ~             |
| 9   | Permission Verification    | find /var/ -type f -perm 0777  | ✓             |

Legend: ✓ Completed | ~ Attempted/In Progress

## Topics & Skills Covered

| <b>Category</b>         | <b>Skills Demonstrated</b>                                                  |
|-------------------------|-----------------------------------------------------------------------------|
| Find Command Mastery    | find with -mmin, -type, -name, -perm flags<br>Recursive directory traversal |
| Permission Basics       | chmod with g-w (remove group write)<br>Permission filtering with -perm /g+w |
| Advanced Filtering      | Combining multiple find conditions<br>Excluding results with ! operator     |
| File Output Redirection | Saving find results to files (>)<br>File management with sudo cp            |
| Permission Systems      | Octal notation (0777)<br>Special permissions (setuid, setgid, sticky)       |

## Key Learnings & Commands

**Ex. 1 | File Search by Time:** Used find /dev/ -mmin -5 to locate files modified in the last 5 minutes. Critical for troubleshooting recent system changes.

**Ex. 2 | Permission Removal:** Applied chmod g-w to remove write permissions from group. Foundational for secure file access control.

**Ex. 3 | Complex Permission Filtering:** Combined find with -perm /g+w and ! (negation) operator to locate files with specific permission criteria. Advanced troubleshooting skill.

**Ex. 6-7 | File Search & Management:** Located files using find with -type and -name filters, then copied with proper permissions using cp. Essential for file administration.

**Ex. 9 | Permission Verification:** Searched for files with specific octal permissions (0777) to audit system security. Critical for compliance and security hardening.

## Next Steps for Advancement

| <b>Skill Gap</b>          | <b>Recommendation</b>                                             |
|---------------------------|-------------------------------------------------------------------|
| Special Permissions       | Master setuid, setgid, sticky bit (chmod with symbolic notation)  |
| Directory Operations      | Practice finding directories only (-type d) with complex criteria |
| Size-based Searches       | Understand find -size syntax variations (k, M, G for units)       |
| Permission Octal Notation | Deep dive into 4-digit octal (setuid=4, setgid=2, sticky=1)       |